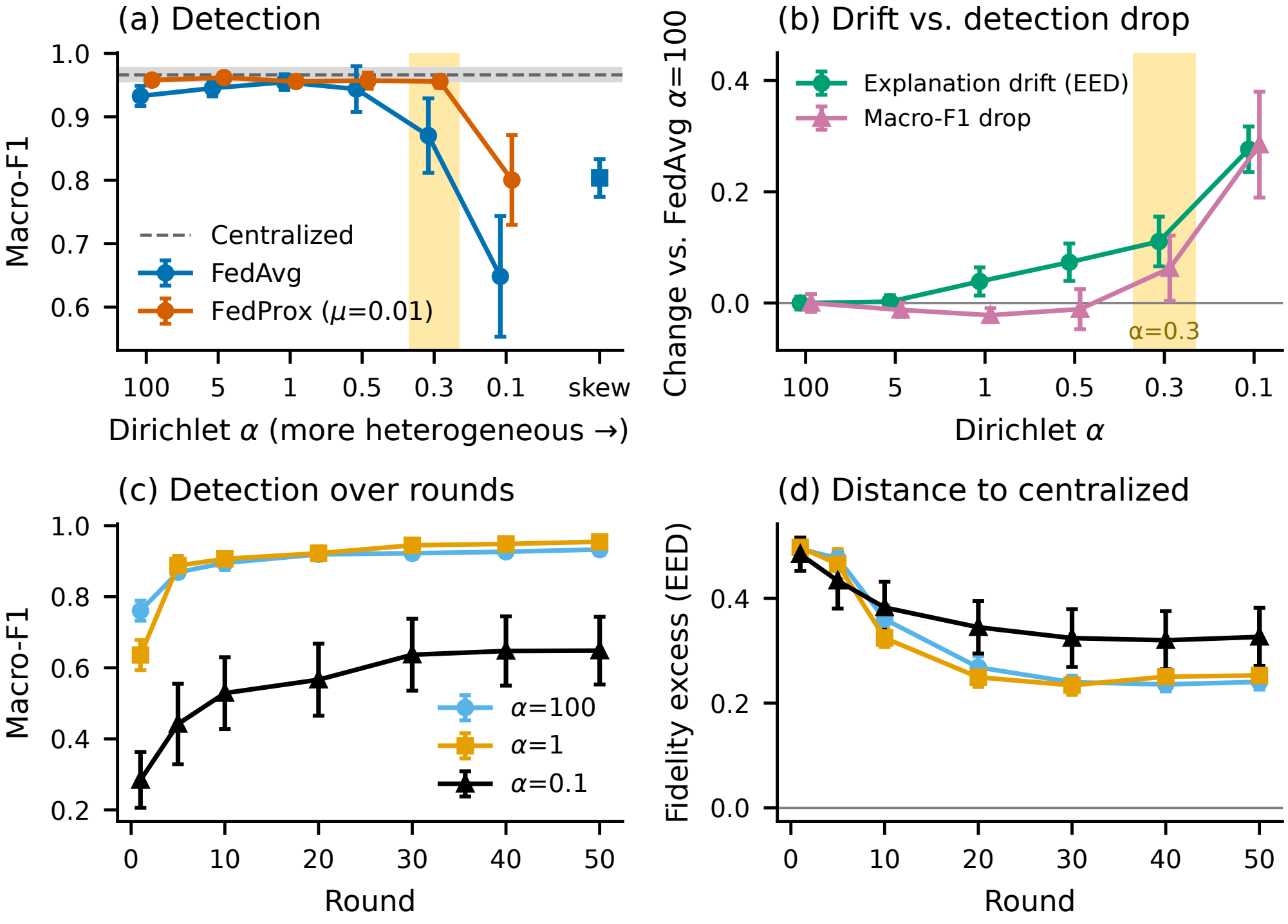
(a) Detection
Macro-F1
1.0
0.9
0.8
0.7
0.6
Centralized
FedAvg
FedProx (μ=0.01)
100
5
1
0.5
0.3
0.1
skew
Dirichlet α (more heterogeneous →)
(b) Drift vs. detection drop
Change vs. FedAvg α=100
0.4
0.2
0.0
Explanation drift (EED)
Macro-F1 drop
α=0.3
100
5
1
0.5
0.3
0.1
Dirichlet α
(c) Detection over rounds
Macro-F1
1.0
0.8
0.6
0.4
0.2
α=100
α=1
α=0.1
0
10
20
30
40
50
Round
(d) Distance to centralized
Fidelity excess (EED)
0.4
0.2
0.0
0
10
20
30
40
50
Round